Chapter 29

The Shapley Value

The Shapley value is a popular solution concept in cooperative game theory that provides a unique allocation to a set of players in a coalitional game. Like the Nash bargaining solution, the Shapley value is based on a set of axioms that determine the allocations to different players based on the axioms. The Shapley value faithfully captures the marginal contributions of the players in deciding the allocations. It has found widespread applications in a variety of engineering applications. In this chapter, we present the Shapley axioms and prove the existence and uniqueness of the Shapley value. We present several illustrative examples and also prove key results.

Given a coalitional game, the core may be empty or may be very large or even uncountably infinite. These certainly cause difficulties in getting sharp predictions for the game. The *Shapley value* is a solution concept which is motivated by the need to have a solution concept that would predict a unique expected payoff allocation for every given coalitional game. The Shapley value concept was proposed using an axiomatic approach by Shapley in 1953, as a part of his doctoral dissertation at the Princeton University. Given a game in coalitional form (N, v), we denote the Shapley value by $\phi(N, v) = (\phi_1(N, v), \ldots, \phi_n(N, v))$ where $\phi_i(N, v)$ is the expected payoff to player i. When there is no confusion, we use $\phi(v)$ instead of $\phi(N, v)$, $\phi_i(v)$ instead of $\phi_i(N, v)$, etc. Also, note that $v \in \mathbb{R}^{2^n-1}$, where we have omitted $v(\emptyset)$ which is obviously equal to 0.

The Shapley value tries to capture how coalitional competitive forces influence the possible outcomes of a game. It describes a reasonable or fair way of dividing the gains from cooperation given the strategic realities captured by the characteristic function. It crucially uses the marginal contributions of players as a guiding factor.

Note. In this chapter, as we have done in some earlier chapters, we shall (by slight abuse of notation) denote sets $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 2, 3\}$ simply by 1, 2, 3, 12, 123, respectively. This is purely for the sake of convenience.

Lloyd Shapley can be considered as one of the most influential game theorists of all time. In particular, he is the most important contributor to cooperative game theory. He has made numerous pioneering contributions which have started new areas in game theory. He is a co-recipient with Alvin Roth of the Nobel Prize in Economic Sciences for the year 2012. Many concepts, lemmas and theorems have been named after him. These include, first and foremost, the Shapley value, a popular solution concept in cooperative game theory.

Shapley's pioneering contributions include: (1) Bondareva - Shapley theorem (see previous chapter) which provides a necessary and sufficient condition for the non-emptiness of the core of a coalitional game and which also implies that convex games have non-empty cores (2) Gale - Shapley algorithm [1] which provides the first and perhaps the most used solution to the stable marriage problem (3) Aumann - Shapley pricing that pioneered the pricing of products and services that share resources (4) Shapley - Folkmann lemma which settled the question of convexity of addition of sets (5) Shapley-Shubik power index [2] for determining voting power. Moreover, stochastic games were first proposed by Shapley as early as 1953 [3]. Potential games which are extensively used by researchers these days were proposed by Monderer and Shapley in 1996 [4]. His joint work with Maschler and Peleg on the kernel and the nucleolus is quite path breaking and so is his work with Robert Aumann on non-atomic games and on long-term competition.

Shapley was born in 1923 in Cambridge, Massachusetts and he was a genius in mathematics from an early age. He joined Harvard university for his undergraduate studies and got his A.B. Degree in 1948. During 1943 - 45, he worked for the American Military and as a Sargeant in Army Corps in 1943, he brilliantly broke the Soviet weather code and was decorated with a Bronze star at the age of 20. He started working with Albert Tucker at the Princeton University in 1949 and got his Ph.D. in 1953. The work on what is now called the Shapley value was carried out during this period. The title of his doctoral dissertation was *Additive and Nonadditive Set Functions*. John Nash and Shapley were contemporaries working with the celebrated Albert Tucker. From 1954 to 1981, Shapley spent 27 years at the Rand Corporation as research mathematician and produced many path breaking results. Since 1981, he has been at the University of California, Los Angeles, working in the Department of Economics as well as the Department of Mathematics.

29.1 The Shapley Axioms

Let (N, v) be a game in coalitional form. Let π be a permutation on the set N. Let $(N, \pi v)$ be the coalitional game with the permuted value function given by

$$\pi v(\{\pi(i) : i \in C\}) = v(C), \quad \forall\, C \subseteq N$$

This means that the role of any player $i \in N$, in the game (N, v) is essentially the same as the role of player $\pi(i)$ in $(N, \pi v)$.

Example 29.1. Suppose $N = \{1, 2, 3\}$. Consider the permutation π on N de-

fined by $\pi(1) = 3; \pi(2) = 1; \pi(3) = 2$. Then the game $(N, \pi v)$ will be the following: $\pi v(1) = v(2)$; $\pi v(2) = v(3)$; $\pi v(3) = v(1)$; $\pi v(12) = v(23)$; $\pi v(23) = v(13)$; $\pi v(13) = v(12)$; $\pi v(123) = v(123)$. □

Shapley proposed three axioms to describe the desirable properties that we would expect a good solution concept to satisfy: (1) Axiom 1 : Symmetry; (2) Axiom 2 : Linearity; (3) Axiom 3 : Carrier.

Axiom 1: Symmetry

For any $v \in \mathbb{R}^{2^n-1}$, any permutation π on N, and any player $i \in N$, the symmetry axiom states that

$$\phi_{\pi(i)}(\pi v) = \phi_i(v)$$

Informally, the Shapley value of a player relabeled by a permutation, under the permuted value function is the same as the Shapley value of the original player under the original value function. This axiom implies that only the role of a player in the game should matter. The labels or specific names used in N are irrelevant.

Axiom 2: Linearity

Let (N, v) and (N, w) be any two coalitional games. Suppose $p \in [0, 1]$. Define a new coalitional game $(N,\ pv + (1-p)w)$ as follows.

$$(pv + (1-p)w)(C) = pv(C) + (1-p)w(C) \ \ \forall C \subseteq N$$

Axiom 2 states that, given any two coalitional games (N, v) and (N, w), any number $p \in [0, 1]$, and any player $i \in N$,

$$\phi_i(pv + (1-p)w) = p\phi_i(v) + (1-p)\phi_i(w)$$

In other words, the Shapley value of a player for a convex combination of coalitional games is the convex combination of Shapley values of the player in the individual games. This axiom asserts that the expected payoff to each player is the same before resolution of uncertainty and after resolution of uncertainty.

Axiom 3: Carrier

Definition 29.1 (Carrier). *A coalition D is said to be a* carrier *of a coalitional game (N, v) if*

$$v(C \cap D) = v(C) \quad \forall C \subseteq N$$

If D is a carrier and $i \notin D$, then

$$v(\{i\}) = v(\{i\} \cap D) = v(\phi) = 0$$

If D is a carrier of (N, v), all players $j \in N \setminus D$ are called *dummies* in (N, v) because their entry into any coalition cannot change the worth of the coalition. Also, for any $C \subseteq N$ and $i \notin D$,

$$v(C \cup \{i\}) = v((C \cup \{i\}) \cap D) = v(C \cap D) = v(C)$$

Intuitively, D includes all influential players (it might also include non-influential players but it will not exclude any influential player). If D is a carrier and $i \in N$, then for any $C \subseteq N$,

$$v(C) = v(C \cap D) = v(C \cap (D \cup \{i\}))$$

Hence, $D \cup \{i\}$ for any $i \in N$ is also a carrier. In fact, the set N is always a carrier. This means that $v(D) = v(N)$. This can also be seen from

$$v(D) = v(D \cap N) = v(N)$$

It is however possible that no proper subset of N is a carrier.

The *Carrier Axiom* (Axiom 3) states that, for any $v \in \mathbb{R}^{2^n-1}$ and any coalition D that is a carrier of (N, v),

$$\sum_{i \in D} \phi_i(v) = v(D) = v(N)$$

The carrier axiom immediately implies that

$$\phi_i(v) = 0 \ \ \forall\, i \notin D$$

$$\sum_{i \in N} \phi_i(v) = v(N)$$

This axiom asserts that the players in a carrier set should divide their joint worth (which is equal to the worth of the grand coalition) among themselves. This means the dummies are allocated nothing. The above expression also illustrates a key fact that the Shapley value always divides the worth of the grand coalition among the players of the game. This means that Shapley value implicitly assumes the formation of the grand coalition (however some of these players may be dummy players who do not get anything allocated).

Note. Throughout the rest of this chapter, we will denote, for the sake of convenience, the set $N \setminus \{i\}$ by the notation $N - i$.

29.2 Shapley's Theorem and Examples

With the above three axioms in place, we are now in a position to state the celebrated result due to Shapley [5].

Theorem 29.1. *There is exactly one mapping $\phi : \mathbb{R}^{2^n-1} \to \mathbb{R}^n$ that satisfies Axiom 1, Axiom 2, and Axiom 3. This mapping satisfies: $\forall i \in N$, $\forall v \in \mathbb{R}^{2^n-1}$,*

$$\phi_i(v) = \sum_{C \subseteq N-i} \frac{|C|!\ \ (n - |C| - 1)!}{n!} \ \ \{v(C \cup \{i\}) - v(C)\}$$

The term $\frac{|C|!\ (n-|C|-1)!}{n!}$ can be interpreted as the probability that in any permutation, the members of C are ahead of a distinguished player i. The term $v(C \cup \{i\}) - v(C)$ gives the marginal contribution of player i to the worth of the coalition C. Thus the above formula for $\phi_i(v)$ gives the expected contribution of player i to the worth of any coalition.

Suppose there is a collection of n resources and each resource is useful in its own way towards executing a certain service. Suppose $v(N)$ is the total value that this collection of resources would create if all the resources are deployed for the accomplishment of the service. Let us focus on a certain resource, say resource i. Now, this resource will make a marginal contribution to every subset C of $N - i$ when it is included to the set C. We can choose the set C in $(|C|!\ \ |n-|C|-1|!)$ ways and when this is divided by $n!$, we obtain the probability of choosing a particular subset C. Thus the Shapley value of resource i is the average marginal contribution that resource i will make to any arbitrary coalition of resources that is a subset of $N - i$.

Alternatively, imagine that there is committee room where an important discussion is in progress and only one person can enter the committee room at a time. Suppose the committee room currently consists of individuals belonging to a certain coalition C with the rest of the individuals outside the committee room. An individual $i \notin C$ entering the committee room will contribute in a certain way to the ongoing discussion in the committee room. The contribution of i will pretty much depend on how influential, powerful, and articulate the individual i is. If the committee room is equally likely to have any coalition that is a subset of $N - i$, then the average contribution i would make to any arbitrary such coalition would be the Shapley value of i.

Example 29.2 (Divide the Dollar Game). First, let us consider Version 3 (Example 27.3). Recall that $N = \{1, 2, 3\}$; $v(1) = v(2) = v(3) = v(23) = 0$; $v(12) = v(13) = v(123) = 300$. The Shapley value expression for $\phi_1(v)$ would be:

$$\phi_1(v) = \frac{2}{6}(v(1) - v(\emptyset)) + \frac{1}{6}(v(12) - v(2)) + \frac{1}{6}(v(13) - v(3)) + \frac{2}{6}(v(123) - v(23))$$

Similarly,

$$\phi_2(v) = \frac{2}{6}(v(2) - v(\emptyset)) + \frac{1}{6}(v(12) - v(1)) + \frac{1}{6}(v(23) - v(3)) + \frac{2}{6}(v(123) - v(13))$$

$$\phi_3(v) = \frac{2}{6}(v(3) - v(\emptyset)) + \frac{1}{6}(v(13) - v(1)) + \frac{1}{6}(v(23) - v(2)) + \frac{2}{6}(v(123) - v(12))$$

It can be easily seen that

$$\phi_1(v) = 200; \ \ \phi_2(v) = 50; \ \ \phi_3(v) = 50$$

Also, it can be easily verified from the above expressions that

$$\phi_1(v) + \ \ \phi_2(v) + \ \ \phi_3(v) = v(123)$$

It can also be verified easily that $\phi = (100, 100, 100)$ for Version 1 (Example 27.1) and Version 4 (Example 27.4) of the game, while $\phi = (150, 150, 0)$ for Version 2 (Example 27.2) of the game. □

Example 29.3 (Minimum Spanning Tree Game). Consider the minimum spanning tree game considered in Example 27.7. Recall that $N = \{1, 2, 3\}$ and

$$v(\emptyset) = 0;\ v(1) = 5;\ v(2) = 9;\ v(3) = 7$$

$$v(12) = 15;\ v(13) = 12;\ v(23) = 17; v(123) = 23.$$

Following the expressions in the previous example, it can be shown that

$$\phi_1(v) = 5.5;\ \ \phi_2(v) = 10;\ \ \phi_3(v) = 7.5$$

Notice that the payoff allocation to player 2 is the highest which is clearly justified. Also, it is logical that player 1 gets the least allocation while player 3's allocation is greater than that of player 1. Also, note that this allocation belongs to the core of the game. □

Example 29.4 (Glove Market). Consider the glove market game (Example 28.4) with $2,000,001$ players, having one $1,00,000$ left glove suppliers and $1,000,001$ right glove suppliers. We have seen that the core of this game is a singleton. Here, the Shapley value of any right-glove supplier is the probability that he would find in any coalition more left glove suppliers than right-glove suppliers. Here the Shapley value for each right-glove supplier is 0.499557 and for each left glove supplier is 0.500443. Note that the Shapley value is able to capture the effect of coalitional forces better than the core. Also note that the Shapley value does not belong to the core. □

Example 29.5 (Apex Game). In this game, there are five players: $N = \{1, 2, 3, 4, 5\}$. Player 1 is called the big player and the other players are called small players. The big player with one or more small players can earn a worth of 1. The four small players together can also earn 1.

$$\begin{aligned} v(C) &= 1 \text{ if } 1 \in C \text{ and } |C| \geq 2 \\ &= 1 \text{ if } |C| \geq 4 \\ &= 0 \text{ otherwise} \end{aligned}$$

The Shapley value is

$$\phi(v) = \left(\frac{3}{5}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}\right)$$

Note that the core of this game is empty. □

Example 29.6 (A Logistics Game). Figure 29.1 shows a logistics network that provides connectivity between two important cities S and T. There are five hubs A, B, C, D, E which are intermediate points from S to T. The transportation is provided by service

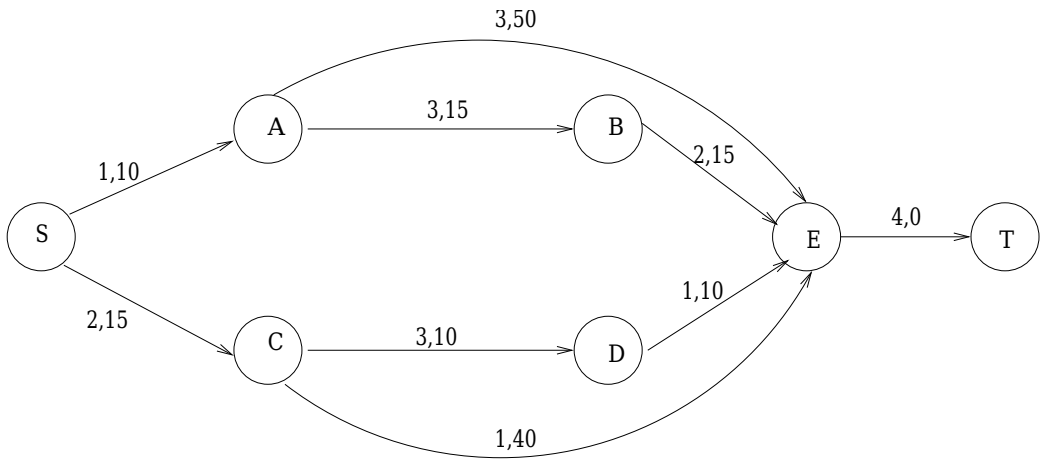

Fig. 29.1: A logistics network

providers 1, 2, 3, 4. We have already seen this example in Chapter 27, where we formulated this as a TU game with $N = \{1, 2, 3, 4\}$ and with characteristic function

$$v(1) = v(2) = v(3) = v(4) = 0$$

$$v(12) = v(13) = v(14) = v(23) = v(24) = v(34) = v(234) = v(123) = 0$$

$$v(134) = 40; \;\; v(124) = 45; \;\; v(1234) = 65$$

It can be shown that the Shapley values here are given by $\phi(v) = (20, 20, 5, 20)$. This can be shown to belong to the core of this game. □

29.3 Proof of the Shapley Theorem

The proof proceeds in two parts. In Part 1, we show that the formula for $\phi_i(v)$ satisfies all the three axioms. In Part 2, we show that there exists a unique mapping ϕ that satisfies all the three axioms.

Proof of Part 1

Symmetry

Observe that in the formula for $\phi_i(v)$, what only matters about a coalition is whether it contains i and the number of players it contains. Thus relabeling does not affect the value in any way. This observation clearly shows that symmetry (Axiom 1) is satisfied.

Linearity

To show linearity, we have to show that

$$\phi_i(pv + (1-p)w) = p\phi_i(v) + (1-p)\phi_i(w)$$

We have, by definition, for any $p \in [0,1]$,

$$(pv + (1-p)w)(C) = pv(C) + (1-p)w(C) \; \forall C \subseteq N \tag{29.1}$$

Note that $\phi_i(pv + (1-p)w)$ is equal to

$$\sum_{C \subseteq N-i} \frac{|C|! \;\; (n-|C|-1)!}{n!} \;\; ((pv + (1-p)w)(C \cup \{i\}) - (pv + (1-p)w)(C))$$

By expanding this and applying equation (29.1), linearity can be established.

Carrier Axiom

Suppose D is a carrier of (N, v). Then, we know that

$$v(C \cap D) = v(C) \; \forall C \subseteq N$$

We have also seen that

$$v(\{i\}) = 0 \quad \forall \;\; i \in N \setminus D \quad \text{and} \quad v(D) = v(N)$$

If we take a look at the formula for the Shapley value, it is very clear that

$$\phi_i(v) = 0 \; \forall i \in N \setminus D$$

since

$$v(C \cup \{i\}) = v((C \cup \{i\}) \cap D) = v(C \cap D) = v(C)$$

We have to show for all carriers D that

$$\sum_{i \in D} \phi_i(v) = v(D)$$

Substituting the formula for $\phi_i(v)$ from the Shapley theorem and simplifying, we can show that

$$\sum_{i \in N} \phi_i(v) = v(N)$$

Since D is a carrier, we have $v(D) = v(N)$, hence the carrier axiom follows.

Proof of Part 2

Here we show that there exists a unique mapping ϕ that satisfies the three axioms. First we prove that the mapping ϕ is a linear transformation by making the following observations.

- Let (N, z) be the coalitional game that assigns worth zero to every coalition, that is, $z(C) = 0 \ \forall C \subseteq N$. Then Axiom 3 (carrier axiom) implies that

$$\phi_i(z) = 0 \ \forall i \in N \tag{29.2}$$

- From Axiom 2, we have

$$\phi_i(pv + (1-p)w) = p\phi_i(v) + (1-p)\phi_i(w)$$

  Choosing $w = z$ in the above, we get

$$\phi_i(pv) = p\phi_i(v) \ \ \forall i \in N \tag{29.3}$$

Equations (29.2) and (29.3) together with the linearity axiom imply that ϕ is a linear transformation.

Suppose $L(N)$ denotes the set of all non-empty subsets of N. Clearly, $|L(N)| = 2^n - 1$ and hence $\mathbb{R}^{|L(N)|}$ is a $(2^n - 1)$-dimensional vector space. Let $D \subseteq N$ be any coalition. Define for D, a coalitional game (N, w_D):

$$\begin{aligned} w_D(C) &= 1 \ \text{ if } D \subseteq C \\ &= 0 \ \text{ otherwise} \end{aligned}$$

This implies that a coalition C has worth 1 in w_D if it contains all the players in D and has worth zero otherwise. The game (N, w_D) is called the simple D-carrier game. To get a feel for this game, we discuss a simple example before resuming the proof.

Example 29.7 (D-carrier game). Let $N = \{1, 2, 3\}$. Let w_1, w_2, w_3, w_{12}, w_{13}, w_{23}, and w_{123} be the characteristic functions of the D-carrier games corresponding to the coalitions $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$, respectively. For instance, w_{12} would be:

$$w_{12}(1) = 0; w_{12}(2) = 0; w_{12}(3) = 0; w_{12}(12) = 1; w_{12}(13) = 0; w_{12}(23) = 0; w_{12}(123) = 1$$

We can write all the w values as follows. Note that each one is a 7-tuple.

$$\begin{aligned} w_1 &= (1, 0, 0, 1, 1, 0, 1) \\ w_2 &= (0, 1, 0, 1, 0, 1, 1) \\ w_3 &= (0, 0, 1, 0, 1, 1, 1) \\ w_{12} &= (0, 0, 0, 1, 0, 0, 1) \\ w_{13} &= (0, 0, 0, 0, 1, 0, 1) \\ w_{23} &= (0, 0, 0, 0, 0, 1, 1) \\ w_{123} &= (0, 0, 0, 0, 0, 0, 1) \end{aligned}$$

We will be showing subsequently that these seven vectors is a basis of $\mathbb{R}^7$. $\square$

Continuing the proof of Part 2, note that D is a carrier of (N, w_D) since

$$w_D(C) = w_D(C \cap D) \ \ \forall C \subseteq N$$

In fact, $D \cup \{j\}$ is also a carrier $\forall j \in N$. Also

$$w_D(N) = 1 \text{ since } D \subseteq N$$
$$w_D(D) = 1 \text{ since } D \subseteq D$$

Application of the carrier axiom leads to the following

$$\sum_{i \in D} \phi_i(w_D) = 1 = w_D(D) = w_D(N) \ \text{ and}$$
$$\phi_j(w_D) = 0 \ \ \forall j \notin D$$

The above holds because both D and $D \cup \{j\}$ are carriers of w_D.

Now, by the symmetry axiom, all players in D should get the same payoff. This is because the players in D all contribute equally to the characteristic function w_D. There is no way to distinguish them in any manner, so relabeling can be freely done. This implies:

$$\phi_i(w_D) = \frac{1}{|D|} \ \ \forall i \in D$$
$$\phi_j(w_D) = 0 \ \ \forall j \notin D$$

For each coalition $D \subseteq N$, there is a game (N, w_D) and therefore we have $2^n - 1$ such games. Each game is represented by a characteristic function which is a vector of $2^n - 1$ elements. Thus we have $2^n - 1$ vectors each of size $2^n - 1$. We now show that these are linearly independent in the space $\mathbb{R}^{2^n-1}$. To show linear independence, we have to show that

$$\sum_{D \in L(N)} \lambda_D w_D = (0, 0, \ldots, 0) \Longrightarrow \lambda_D = 0 \ \ \forall D \in L(N)$$

Suppose not. Now,

$$\sum_{D \in L(N)} \lambda_D w_D = (0, 0, \ldots, 0) \ \ \Longrightarrow \ \ \sum_{D \in L(N)} \lambda_D w_D(C) = 0 \ \ \forall C \subseteq N$$

Let $C \subseteq N$ be any coalition of minimal size such that $\lambda_C \neq 0$. Based on the observation that D is either a subset of C or not, we can write:

$$\sum_{D \subseteq C} \lambda_D w_D(C) + \sum_{D \subsetneq C} \lambda_D w_D(C) = 0 \Longrightarrow \ \ \sum_{D \in L(N)} \lambda_D w_D(C) = 0 \ \ \forall C \subseteq N$$

We know that $w_D(C) = 1 \ \ \forall D \subseteq C$ and $w_D(C) = 0$ otherwise (that is, if D is not a subset of C). Therefore, the above implication leads to

$$\sum_{D \subseteq C} \lambda_D = 0$$

Since C is a subset of minimal size for which $\lambda_C \neq 0$, we have that $\lambda_D = 0$ for all subsets $D \subset C$. Thus the above sum is equal to λ_C. This means $\lambda_C = 0$. This is a contradiction! Thus the set $\{w_D : D \in L(N)\}$ is a basis of the space $\mathbb{R}^{2^n-1}$. Since a linear transformation is completely determined by what it does on a basis of the vector space, the mapping ϕ is unique. ■

29.4 Alternative Formulae for Shapley Value

We now describe several alternative ways of developing the Shapley value. Each of these will provide an interesting interpretation for this solution concept.

Approach 1

Given a TU game (N, v), the Shapley value can be written as :

$$\phi_i(v) = \sum_{C \subseteq N-i} \frac{|C|!(n-|C|-1)!}{n!} \{v(N \setminus C) - v(C)\}$$

This is because the coefficients of $\{v(N \setminus C) - v(C)\}$ and $\{v(C \cup \{i\}) - v(C)$ in the above expression are the same. The above expression shows that the value of each player depends only on the difference between the worths of the complementary coalitions. For each pair of complementary coalitions C and $N \setminus C$, the values of players in $N \setminus C$ increase as $v(N \setminus C) - v(C)$ increase while the values of players in C increase as $v(C) - v(N \setminus C)$ increases.

Approach 2

Another formula for Shapley value is:

$$\phi_i(v) = \sum_{\substack{C \subseteq N \\ i \in C}} \frac{(|C|-1)!(n-|C|)!}{n!} \{v(N) - v(N \setminus \{i\})\}$$

It is to be noted that

$$\frac{(|C|-1)!(n-|C|)!}{n!}$$

is the probability that in a random permutation, the coalition C arises as the union of $\{i\}$ with all the predecessors of i.

Approach 3

Let $C \subseteq N$ and $i \notin C$. The marginal contribution of i to C, denoted by $m(C, i)$, is given by:

$$m(C, i) = v(C \cup \{i\}) - v(C)$$

Given any permutation $\pi \in \Pi(N)$, the set of predecessors $P(\pi, i)$ of i in the permutation π is given by:

$$P(\pi, i) = \{j : \pi(j) < \pi(i)\}$$

We have seen that $\phi_i(v)$ is the average marginal contribution of i to any coalition of N assuming all orderings are equally likely. Another way of interpreting this is that

$\phi_i(v)$ is the average marginal contribution of i to the set of his predecessors where the average taken over all permutations equally likely. We therefore get

$$\phi_i(v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} m(P(\pi, i), i); \ \forall i \in N. \tag{29.4}$$

Similarly, if

$$S(\pi, i) = \{j : \pi(i) < \pi(j)\}$$

denotes the set of all players who are successors of i in the permutation π, we also get

$$\phi_i(v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} m(S(\pi, i), i); \ \forall i \in N. \tag{29.5}$$

One can verify that (29.4) and (29.5) independently satisfy the expression for the Shapley value.

29.5 Shapley Value of Convex Games

In Chapter 27, we have already defined convex games. In Chapter 28, we have also shown that the core of any convex game is non-empty. We now show that the Shapley value of a convex game belongs to the core of the game. Note in general that the Shapley value need not be a member of the core even if the core is non-empty.

Theorem 29.2. *Let (N, v) be a convex game. Then $\phi(v) \in \mathbb{C}(N, v)$.*

Proof: Let $\Pi(N)$ as usual denote the collection of all permutation of the set N. Suppose $\pi \in \Pi(N)$ is any permutation. Define the allocation $y^\pi \in \mathbb{R}^n$ as follows.

$$y^\pi = (m(\pi, 1), m(\pi, 2), \ldots, m(\pi, n)).$$

We have seen in Chapter 28 (Proposition 28.2) that

$$y^\pi \in \mathbb{C}(N, v) \ \ \forall \pi \in \Pi(N).$$

Furthermore, we have already seen that

$$\phi_i(v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} m(P(\pi, i), i)$$

Thus $\phi(v)$ is a convex combination of all vectors y^π; $\pi \in \Pi(N)$. Since the core $\mathbb{C}(N, v)$ is a convex set, it follows immediately that $\phi(v) \in \mathbb{C}(N, v)$. ■

The above result is a useful one. If we are able to model a physical situation as a convex game, then the Shapley value has the added characteristic that it is also a stable allocation since it belongs to the core. Thus for convex games, the Shapley value provides an excellent solution concept.

Example 29.8 (Shapley value of MST profit game). Consider the MST profit game with $N = \{1, 2, 3\}$ corresponding to the minimum spanning tree game discussed in Example 27.7:

$$v(1) = 5;\ v(2) = 9;\ v(3) = 7;\ v(12) = 15;\ v(13) = 12;\ v(23) = 17;\ v(123) = 23$$

This is a convex game and we have shown that the core is given by

$$\{(x_1, x_2, x_3) :\ 5 \leq x_1 \leq 6;\ 9 \leq x_2 \leq 11;\ 7 \leq x_3 \leq 8; x_1 + x_2 + x_3 = 23\}$$

The Shapley value can be shown to be given by $(5.5, 10, 7.5)$ which clearly belongs to the core. It is logical to see that player 2 gets the highest allocation and player 1 gets the lowest allocation in the Shapley value. □

29.6 Shapley - Shubik Power Index

The Shapley - Shubik power index is an apt solution concept used in the context of a decision making process to determine the power of each player in the process. A decision making process involving a set of players $N = \{1, \ldots, n\}$ can be modeled as a simple, monotonic TU game. Recall that a TU game (N, v) is monotonic if

$$v(C) \leq v(D)\ \ \forall C \subseteq D \subseteq N.$$

The game (N, v) is said to be a *simple game* if $v(C)$ is either 0 or 1 for any coalition $C \subseteq N$. The Shapley - Shubik power index of a simple, monotonic TU game is simply the Shapley value of the game. Given a player $i \in N$, define the set

$$Q_i(N, v) = \{C \subseteq N : v(C \cup \{i\}) = 1 \text{ and } v(C) = 0\}$$

That is, $Q_i(N, v)$ consists of all coalitions such that the coalition changes over from a losing coalition to a winning coalition when player i is inducted into the coalition. It can be shown that the Shapley value of simple monotonic games (and hence the Shapley - Shubik power index) is given by

$$\phi_i(v) = \sum_{C \in Q_i(N,v)} \frac{|C|!(n - |C| - 1)!}{n!}$$

Example 29.9 (Shapley - Shubik Power Index). Consider the following three player simple TU game which can be easily shown to be monotonic:

$$v(1) = 1;\ v(2) = v(3) = 0;\ v(12) = v(13) = v(23) = v(123) = 1.$$

Note for this game that

$$Q_1(N, v) = \{\emptyset, \{2\}, \{3\}\};\ Q_2(N, v) = \{\{3\}\};\ Q_3(N, v) = \{\{2\}\}$$

Applying the formula above, we get the Shapley - Shubik power index as $(\frac{2}{3}, \frac{1}{6}, \frac{1}{6})$. This is in fact the Shapley value of the given game. Note that players 2 and 3 have the same index due to symmetry. Player 1 is four times as powerful as player 2 or player 3. □

29.7 Banzhaf Index

Like the Shapley - Shubik power index, the Banzhaf index (often called the Banzhaf power index) also measures the power of a player in a coalitional game. Suppose (N, v) is a TU game and recall the expression for computing the Shapley value (Equation 29.4):

$$\phi_i(N, v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} v(P(\pi, i) \cup \{i\}) - v(P(\pi, i)) \ \ \forall i \in N$$

In the above expression, the average is taken over all permutations which are assumed to be equally likely. On the other hand, consider the following expression

$$\psi_i(N, v) = \frac{1}{2^{n-1}} \sum_{C \subseteq N-i} (v(C \cup \{i\}) - v(C)) \ \ \forall i \in N$$

In the above expression, the average is taken over all coalitions assuming that all coalitions are equally likely. The index $\psi_i(N, v)$ is called the *Banzhaf Index* of player i. This signifies the average marginal contribution of the player i to any coalition under the assumption that all coalitions are equally likely. The following example shows that Banzhaf index is not the same as the Shapley value.

Example 29.10 (Banzhaf Index). Consider the so called *unanimity game* which is any TU game (N, v) such that $v(N) = 1$ and $v(C) = 0 \ \forall C \subset N$. The Shapley value for this game turns out to be

$$\phi_i(N, v) = \frac{1}{n} \ \forall i \in N$$

The Banzhaf index for this game can be shown to be

$$\psi_i(N, v) = \frac{1}{2^{n-1}} \ \forall i \in N$$

Note from the above that the sum of Banzhaf indices for all the players is not equal to $v(N) = 1$. Therefore the Banzhaf does not satisfy Pareto optimality which the Shapley value satisfies. In fact, except Pareto optimality, Banzhaf index satisfies all other axiomatic properties of the Shapley value. □

Several variants of the Banzhaf power index have been proposed in the literature. The paper by Dubey and Shapley [6] may be consulted for more details.

29.8 Summary and References

The Shapley value of a TU game is a unique allocation vector that satisfies three properties axiomatized by Shapley, namely (1) symmetry (2) linearity and (3) the carrier property.

- The Shapley value (in superadditive games) distributes the value of the grand coalition among the players in a way that reflects the marginal contribution of

each node to the game on an average. In this sense, Shapley value captures a fair distribution of total value among individual players. However, the Shapley value implicitly assumes that the grand coalition forms. The carrier property ensures that dummy players get a zero allocation while the other players get non-zero allocations.

- There are many alternate formulations of Shapley value and the chapter has presented several different formulations.
- The Shapley value may or may not belong to the core (if the core is non-empty). This means that the Shapley value, though a fair allocation, may not be a sustainable allocation unlike any allocation that belongs to the core.
- The core of a convex game is non-empty and moreover, the Shapley value of a convex game always belongs to the core.
- The Shapley-Shubik power index is defined for simple, monotonic games that represent a decision making process. This index is obtained by computing the Shapley value of the game. This index captures the power of different players in the decision making process and can be computed efficiently due to the special structure of games to which it is applicable.
- The Banzhaf index is defined as the average marginal contribution of the player i to any coalition under the assumption that all coalitions are equally likely. The Shapley value computation assumes that all permutations are equally likely while the Banzhaf index assumes that all coalitions are equally likely. The Banzhaf index does not satisfy Pareto optimality but satisfies all other axiomatic properties of the Shapley value.

The material discussed in this chapter draws upon mainly from the book by Myerson [7]. The recent book by Maschler, Solan, and Zamir [8] is an excellent source for further reading. The volume edited by Roth [9] embodies a comprehensive account of Shapley value and extensions until 1988.

Computation of Shapley value is a hard problem and clearly it has exponential time complexity. The book by Chalkiadakis, Elkind, and Wooldridge [10] has pointers to several references on this important topic. In special cases such as convex games (where the Shapley value is in fact the center of gravity of the core), the computation is somewhat easier.

The Shapley value is extremely useful in practical applications. Two such applications are efficient methods for clustering [11] and determining most influential nodes in social networks [12].

The Shapley-Shubik power index was introduced in [2] and the Banzhaf index is discussed in [13]. For more details about the Banzhaf index, the paper by Dubey and Shapley [6] can be consulted.

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29.9 Exercises

(1) Show using the expression for Shapley value that the sum of Shapley values of all players will be equal to the value of the grand coalition.

(2) Suppose (N, v) is a TU game and we define a unique imputation as follows.

$$\xi_i(N, v) = v(\{i\})\ \forall i \in N.$$

Which of the Shapley axioms does the above satisfy and which of the Shapley axioms does it violate?

(3) Suppose (N, v) is a TU game and we define a unique imputation as follows.

$$\xi_i(N, v) = v(\{1, \ldots, i-1, i\} - v(\{1, \ldots, i-1\})\ \forall i \in N.$$

Which of the Shapley axioms does the above satisfy and which of the Shapley axioms does it violate.

(4) Consider a three person superadditive game with $v(1) = v(2) = v(3) = 0; v(12) = a; v(13) = b; v(23) = c; v(123) = d$ where $0 \leq a, b, c \leq d$. Compute the Shapley value for this game.

(5) Consider the following characteristic form game with three players. $v(1) = v(2) = v(3) = 0; v(12) = a; v(13) = b; v(23) = c; v(123) = 1$. Assume that $0 \leq a, b, c \leq 1$.

(a) Find the conditions under which the core is non-empty.
(b) Compute the Shapley value.
(c) Assuming the core is non-empty, does the Shapley value belong to the core? Under what conditions will the Shapley value belong to the core of this game.

(6) Let us consider a version of divide the dollar problem with 4 players and total worth equal to 400. Suppose that any coalition with three or more players will be able to achieve the total worth. Also, a coalition with two players will be able to achieve the total worth only if player 1 is a part of the two player coalition. Set up a characteristic function for this TU game and compute the Shapley value.

(7) There are four players $\{1, 2, 3, 4\}$ who are interested in a wealth of 400 (real number). Any coalition containing at least two players and having player 1 would be able to achieve the total wealth of 400. Similarly, any coalition containing at least three players and containing player 2 also would be able to achieve the total wealth of 400. Set up a characteristic form game for this situation and compute the Shapley value.

(8) Give simple examples of two player games where

- The core is empty and Nash Bargaining Solution (NBS) = Shapley Value (SV)
- The core is empty and NBS is not the same as SV
- The core is non-empty and NBS = SV
- The core is non-empty and NBS is not the same as SV

(9) Give simple examples of TU games where

- The core is non-empty and the Shapley value belongs to the core
- The core is non-empty and the Shapley value does not belong to the core

(10) A TU game (N, v) is said to be additive if

$$v(C) = \sum_{i \in C} v(\{i\}) \; \forall C \subseteq N.$$

Determine the Shapley value of such a game in closed form.

(11) Given a TU game (N, v), define the dual game (N, w) by

$$w(C) = v(N) - v(N \setminus C) \; \forall C \subseteq N.$$

Show that the dual of the dual game is the original game (primal game) itself. Also show that the Shapley values of the primal game and the dual game are identical.

(12) Consider the following three player simple TU game which can be easily shown to be monotonic:

$$v(1) = v(2) = v(3) = v(23) = 0;\ v(12) = v(13) = v(123) = 1.$$

In fact, the above game is modeled on the lines of version 3 of the divide the dollar game. Determine the Shapley-Shubik power index of this game.

(13) Consider a TU game with four players where

$$v(12) = V(13) = v(123) = v(134) = v(124) = v(234) = v(1234) = 1.$$

The characteristic function takes zero value for the rest of the coalitions. Is this game monotonic? If yes, compute the Shapley - Shubik power index for this game.

(14) Is the voting game discussed in Chapter 27 (Example 27.6) monotonic? If yes, compute the Shapley - Shubik power index for this game.

(15) Is the apex game monotonic? If yes, compute the Shapley - Shubik power index for this game.

(16) **Programming Assignment**. Undertake a survey of computational methods available for Shapley value computation. Implement efficient algorithms for computing Shapley value.

(17) **Programming Assignment**. Design and implement a program for finding out if a given TU game is a convex game. Implement efficient algorithms to compute Shapley value of convex games.